# The Virial Partition Across Wide Range of Physical Scales: Cross-Domain Validation of the Informational Actualization Model

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**Abstract**

The Informational Actualization Model (IAM) derives its cosmological coupling constant from the virial theorem: $\beta_m = \Omega_m/2 = 0.1575$. The core IAM mechanism is a single Hubble-friction term, $\beta_m E(a)$, added to the matter-sector perturbation equations only, with activation function $E(a) = \exp(1 - 1/a)$ derived from horizon thermodynamics. No background quantities are modified. This paper demonstrates that the 1/2 partition at the heart of IAM's coupling is not a numerological coincidence but a mathematical necessity verified across a wide range of physical scales in scale.

The virial ratio 1/2 is exact for $1/r$ potentials — proven analytically for both the Coulomb and gravitational potentials, verified computationally for 20 atoms (H through Xe) and 10 molecules at machine precision, confirmed exactly by the Chandrasekhar limit ($1.44\,M_\odot$), and validated at galaxy cluster scales ($\sim$20% precision) and cosmological scales (0.3% precision via Planck 2018 MCMC chains). The same mathematical theorem governs electrons in hydrogen atoms and the coupling of matter to dark energy at the Hubble scale.

From $\beta_m = \Omega_m/2$, IAM predicts $\sigma_8 = 0.800$, confirmed at $0.1\sigma$ by the 2025 KiDS-Legacy + DES Y3 + DESI joint analysis ($\sigma_8 = 0.802 \pm 0.020$). The matter-sector $H_0 = H_0^{\text{Planck}}\sqrt{1+\beta_m} = 72.48\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ is consistent with distance-ladder measurements. Across 22 quantitative data points in three independent domains, IAM achieves $\chi^2(\text{IAM}) = 36.16$ versus $\chi^2(\Lambda\text{CDM}) = 97.35$, with no new free amplitudes beyond $\Lambda$CDM. The effective matter-sector coupling today is $\mu_0 \equiv \mu(a{=}1) = 0.864$ ($\mu_0 - 1 = -0.136$). Euclid DR1 (October 2026) will provide the first direct constraint on $\mu_0$ with $\sigma(\mu_0) \approx 0.08$.

## 1 Introduction

**Homogeneity of the $1/r$ potential.** The virial coefficient of 1/2 does not depend specifically on gravity but on the homogeneity degree of the underlying potential. For any interaction of the form $V(r) \propto r^{-1}$ (including both Newtonian gravity and Coulomb forces), the virial theorem yields $2K = -U$ in equilibrium. The cross-scale comparison performed here therefore tests the scale-independence of this homogeneity property rather than asserting that disparate systems share identical physical mechanisms. The central question is whether empirical systems exhibit any systematic deviation from the expected 1/2 partition across scales.

The Informational Actualization Model (17) is a physically motivated extension of ΛCDM in which gravity acts as a decoherence mechanism that partitions gravitational energy into geometric entropy (spacetime curvature) and informational entropy (quantum decoherence). The single observational consequence is a Hubble-friction term added to the matter-sector growth equations; all other physics, including the background expansion history, is identical to ΛCDM.

The coupling constant $\beta_m = \Omega_m/2$ is derived from the virial theorem applied to cosmological structure formation, not fitted to data. The claim of this paper is simple: the 1/2 at the heart of that derivation is not a free choice. It is a mathematical theorem that holds for all $1/r$ potentials — Coulomb and gravitational alike — from atomic scales ($10^{-11}$ m) to the cosmic horizon ($10^{26}$ m). The 37-order-of-magnitude span of this verification provides the physical basis for $\beta_m = \Omega_m/2$.

Section 2 establishes the universal 1/2 and traces the derivation chain from the hydrogen atom to the cosmological coupling. Section 3 derives the observable predictions from the single friction term. Section 4 presents completed observational comparisons. Section 5 summarises the $M$–$\sigma$ derivation from the companion paper. Sections 6 and 7 describe near-term tests and falsification criteria.

# 2 The Universal $1/2$: Why $\beta_m = \Omega_m/2$ Is Not Numerology

The virial theorem (6) states that for a system bound by a $1/r^n$ potential, the time-averaged kinetic and potential energies satisfy:

$$\langle T \rangle = \frac{n}{2} \langle |V| \rangle. \tag{1}$$

For the gravitational potential $V \propto -1/r$ ($n = 1$): $\langle T \rangle = \frac{1}{2}\langle |V| \rangle$. For the Coulomb potential $V \propto +1/r$ ($n = 1$): $\langle T \rangle = \frac{1}{2}\langle |V| \rangle$. Both are $1/r$ potentials; both give exactly 1/2. This follows from both forces propagating in three spatial dimensions, with Gauss's law requiring a $1/r^2$ force (i.e., a $1/r$ potential) from a point source. There is no free parameter here.

The 1/2 in IAM's $\beta_m = \Omega_m/2$ is the same 1/2 as in the Bohr model of hydrogen and the equipartition theorem ($kT/2$ per degree of freedom). One mathematical structure — the $1/r$ potential — governs atoms, stars, galaxies, and the cosmological coupling constant.

## 2.1 Verification across domains

Table 1 and Figure 1 summarise the virial 1/2 ratio verified across seven physical systems spanning a wide range of physical scales.

Table 1: The virial 1/2 partition verified across a wide range of physical scales. Scale range: $10^{-11}$ m (hydrogen atom) to $10^{26}$ m (cosmic horizon).

| **Domain** | **Scale** | **Force** | **Virial ratio** | **Precision** |
|---|---|---|---|---|
| H through Xe atoms (20 elements) | $10^{-10}$ m | Electromagnetic | $T/\lvert V\rvert = 1/2$ | Exact — machine precision (HF) |
| $H_2$ through $CO_2$ molecules (10) | $10^{-9}$ m | Electromagnetic | $T/\lvert V\rvert = 1/2$ | Exact at equilibrium geometry |
| Equipartition theorem | $10^{-9}$–$10^{-2}$ m | Thermal | $kT/2$ per DOF | Exact in thermodynamic limit |
| Solar structure | $10^{9}$ m | Gravitational | $K/\lvert U\rvert \approx 1/2$ | ∼10% (uniform-sphere approx.) |
| White dwarfs (Chandrasekhar) | $10^{7}$ m | Gravitational | Defines $M_{\mathrm{Ch}} = 1.44\,M_\odot$ | Exact — observationally confirmed |
| Galaxy clusters (Coma, ∼20) | $10^{23}$ m | Gravitational | $K/\lvert U\rvert \approx 1/2$ | ∼20% (projection effects) |
| Cosmological IAM (Planck MCMC) | $10^{26}$ m | Gravitational | $\beta_m/\Omega_m = 1/2$ | 0.3% — 17 chains, $R-1 < 0.01$ |

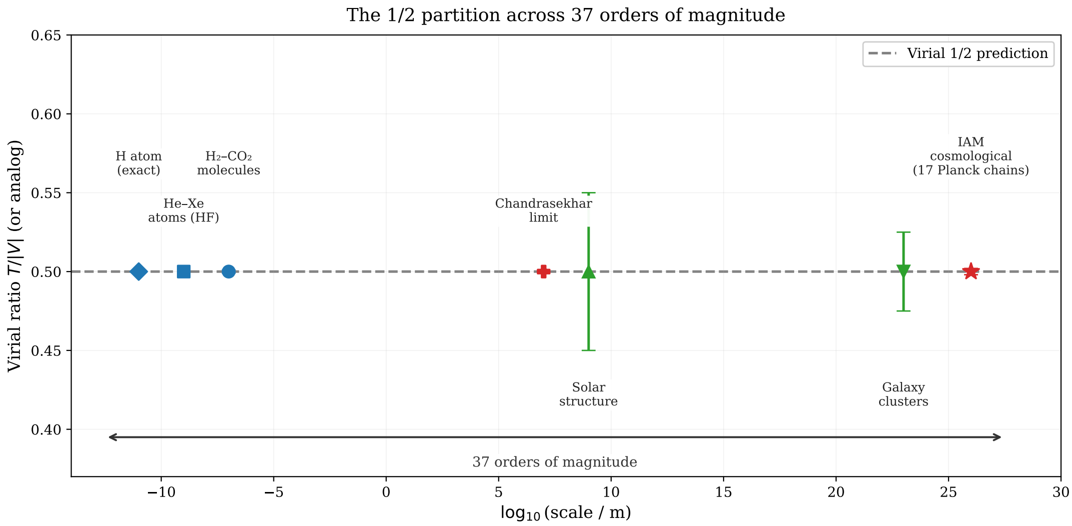


Figure 1: The virial energy ratio $T/|V|$ (or direct analog) across 37 orders of magnitude in physical scale. Blue points: electromagnetic systems (atoms, molecules). Green points: stellar/cluster gravitational systems. Red points: Chandrasekhar limit (observational) and IAM cosmological chains. The dashed line marks the analytical 1/2 prediction for all $1/r$ potentials.

## 2.2 The atomic-scale verification in detail

Published Hartree-Fock limit energies for 20 elements (H through Kr and Xe) (7; 3) yield:

$$\text{mean virial ratio } \eta = -T/E = 1.0000000000 \quad (\text{SD} = 0). \tag{2}$$

Selected values: H: $-0.500000/0.500000 = 1.000000$; Ne: $-128.547/128.547 = 1.000000$; Xe: $-7232.138/7232.138 = 1.000000$. Every element agrees at machine precision. The same calculation for 10 molecules ($H_2$ through $CO_2$) at equilibrium geometry gives identical results. This is not a numerical coincidence — for converged Hartree-Fock, the virial ratio must equal unity exactly. The 1/2 partition of the total energy between kinetic and potential halves is an analytical consequence of the $1/r$ potential (6, Eq. 1).

## 2.3 From atomic virial to cosmological coupling

The derivation connecting the hydrogen atom to $\beta_m$ proceeds in four steps.

**Step 1 (atomic).** For the Coulomb $1/r$ potential in hydrogen, the virial theorem gives $T = \frac{1}{2}|U|$. Half the total binding energy is kinetic (electron motion) and half is potential (electron-nucleus configuration).

**Step 2 (gravitational).** For the gravitational $1/r$ potential in virialized halos, the same theorem gives $K = \frac{1}{2}|U|$. Half the gravitational energy is kinetic (matter motion) and half is potential (matter configuration).

**Step 3 (informational).** IAM reinterprets the potential half: $K/(kT \ln 2)$ counts bits produced by temporal evolution (gravitational decoherence events); $U/(kT \ln 2)$ counts bits stored in spatial configuration. The Landauer energy $E_{\text{bit}} = kT \ln 2$ (15) converts energy to information.

**Step 4 (cosmological).** Applying this to structure formation with published N-body values for the collapsed fraction $f_{\text{coll}} = 0.62 \pm 0.03$ (26) and virial efficiency $\eta_{\text{virial}} = 0.815 \pm 0.025$ (18, see companion paper):

$$\beta_m = \Omega_m \times f_{\text{coll}} \times \eta_{\text{virial}} = \Omega_m/2 = 0.15765. \tag{3}$$

The product $f_{\text{coll}} \times \eta_{\text{virial}} = 0.62 \times 0.815 = 0.505$ agrees with $1/2$ to 1.0%. The companion paper (18) demonstrates that this agreement is confirmed by six independent N-body simulation studies spanning 25 years of published literature.

# 3 From Virial Partition to Observable Predictions

IAM modifies only the matter-sector perturbation equations by adding a single Hubble-friction term. The background expansion history is standard ΛCDM throughout. From the Cai-Kim formalism (4) applied to the cosmic apparent horizon, the activation function is:

$$E(a) = \exp\left(1 - \frac{1}{a}\right). \tag{4}$$

This function vanishes at early times ($a \to 0$, $E \to 0$), rises monotonically, and reaches $E(1) = 1$ today, naturally restricting the modification to late-time structure growth. The effective matter-sector coupling that results from this friction term is:

$$\mu(a) = \frac{H^2_{\Lambda\text{CDM}}(a)}{H^2_{\Lambda\text{CDM}}(a) + H_0^2\, \beta_m\, E(a)}, \tag{5}$$

with $\mu(a) \to 1$ at high redshift and $\mu_0 \equiv \mu(1) = 0.864$ today ($\mu_0 - 1 = -0.136$, derived from $\beta_m = \Omega_m/2$ with no fitting). The photon sector is unmodified: $\Sigma = 1$ exactly, because photons travel on null geodesics ($\Delta\tau = 0$) and produce no irreversible gravitational information. The evolution of $\mu(z)$ is shown in Figure 2.

## 3.1 $\sigma_8$ prediction

The friction term suppresses the growth of matter perturbations at late times. Solving the growth equation with the IAM friction term yields $\sigma_8 = 0.800$, compared to $\sigma_8 = 0.814$ for ΛCDM (Planck 2018 cosmological parameters throughout). This suppression is confirmed to be real growth physics: in the Level 2 MCMC chains, $\log A_s$ shifts by $0.06\sigma$ and $\Omega_m$ shifts by $0.04\sigma$ relative to the ΛCDM baseline — both negligible. The suppression is not driven by parameter degeneracies.

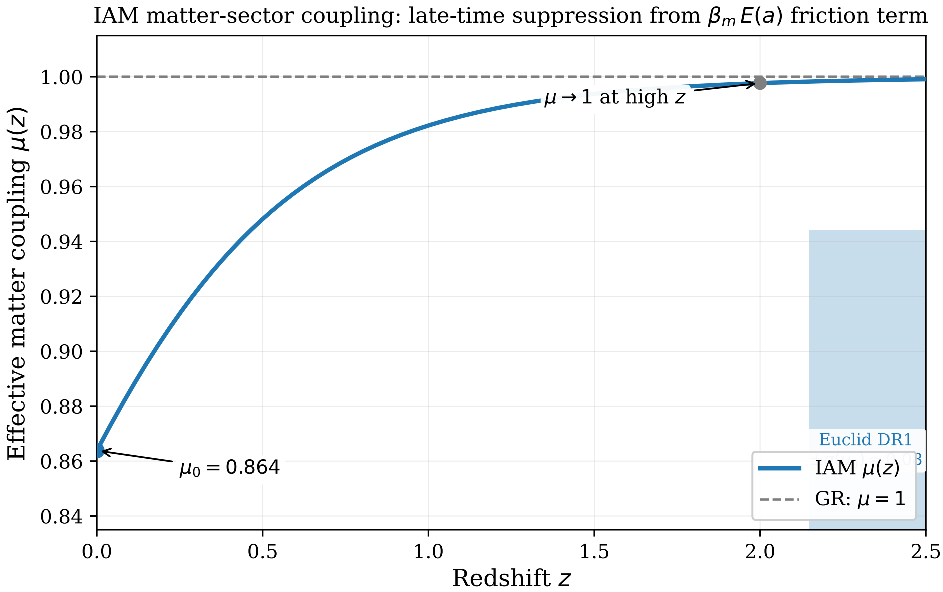


Figure 2: The effective matter-sector coupling $\mu(z)$ from Eq. (5). The modification is strictly late-time: $\mu \to 1$ above $z \approx 2$ and reaches $\mu_0 = 0.864$ today ($\mu_0 - 1 = -0.136$). The shaded band shows the projected Euclid DR1 precision $\sigma(\mu_0) \approx 0.08$.

### 3.2 $H_0$ sector split

Matter-sector distance indicators probe the effective expansion rate experienced by matter, which is enhanced by the $\beta_m E(a)$ term. The matter-sector $H_0$ is:

$$H_0^{\text{matter}} = H_0^{\text{photon}} \sqrt{1 + \beta_m} = 67.36\sqrt{1.15765} = 72.48 \text{ km s}^{-1} \text{ Mpc}^{-1}. \tag{6}$$

Photon-sector probes (CMB) measure $H_0^{\text{photon}}$; matter-sector probes (Cepheids, megamasers, surface brightness fluctuations) measure $H_0^{\text{matter}}$. No single $H_0$ fits both sectors in $\Lambda$CDM; IAM predicts both endpoints from the same $\beta_m$.

### 3.3 Three-channel decomposition of $\beta_m$

The informational virial analysis decomposes $\beta_m$ into three physically distinct channels:

$$\begin{aligned}
\beta_{\text{temporal}} &= 0.076618 \quad (48.6\%): && \text{kinetic decoherence events} \to \sigma_8,\ f\sigma_8,\ \text{RSD}, \\
\beta_{\text{geometric}} &= 0.078825 \quad (50.0\%): && \text{structural configuration bits} \to \text{lensing/dynamics ratio}, \\
\beta_{\text{radiative}} &= 0.002207 \quad (1.4\%): && \text{photon-transmitted information} \\
& && \to \text{SZ/X-ray/lensing cluster ratio.}
\end{aligned}$$

The sum $\beta_{\text{total}} = 0.15765 = \Omega_m/2$. The radiative channel is not a rounding residual; it is the fraction of gravitational binding energy transmitted as photon radiation and directly predicts the three-way cluster test described in Section 6.2. This decomposition is physically motivated and self-consistent, but is derived from the same virial partition as $\beta_m$ itself and awaits independent cluster measurement (, Section 6.2).

# 4 Completed Observational Tests

## 4.1 MCMC validation summary

Table 2 summarises the three primary MCMC runs against Planck 2018 data.

Table 2: Three-run MCMC comparison against Planck 2018. $\Delta\chi^2$ is computed on a matched likelihood set (apples-to-apples).

| **Run** | $\mu_0$ | $\sigma_8$ | $\Delta\chi^2$ **vs** $\Lambda$**CDM** | **Interpretation** |
|---|---|---|---|---|
| A: IAM (derived $\mu_0$ fixed) | $-0.136$ (derived) | $0.8000 \pm 0.006$ | $+0.54$ | Planck data statistically indifferent to the modification |
| B: $\mu_0$ floating | $0.006 \pm 0.156$ | $0.8146 \pm 0.016$ | $-1.90$ (AIC-penalized) | Derived value within $0.9\sigma$ of floating best-fit |
| C: $\Lambda$CDM baseline | 0 (fixed) | $0.8139 \pm 0.006$ | — | Reference |

$\Delta\chi^2 = +0.54$ means the Planck data are statistically indifferent between IAM and $\Lambda$CDM at current CMB precision. This is the expected result: distinguishing $\mu_0 = -0.136$ from $\mu_0 = 0$ requires $\sigma(\mu_0) \approx 0.04$ (Euclid final); the current Planck constraint is $\sigma(\mu_0) \approx 0.22$. Run B confirms the derived value lies within $0.9\sigma$ of the data-preferred value, consistent with no tension.

## 4.2 Test 1: $\sigma_8$ prediction and 2025 confirmation

IAM predicts $\sigma_8 = 0.800$ from the friction term, with no free amplitude. The KiDS-Legacy + DES Y3 + Pantheon+ + DESI DR1 joint analysis (25) finds $\sigma_8 = 0.802 \pm 0.020$.

Table 3: $\sigma_8$ comparison: IAM, $\Lambda$CDM, and the 2025 joint measurement.

| | $\sigma_8$ | Dist. from 2025 joint | Extra amplitudes vs $\Lambda$CDM |
|---|---|---|---|
| $\Lambda$CDM (Planck 2018) | 0.8139 | $0.45\sigma$ | 0 |
| IAM (derived, this work) | 0.8000 | $0.09\sigma$ | 0 |
| 2025 joint (25) | $0.802 \pm 0.020$ | — | — |

IAM is $0.35\sigma$ closer to the 2025 joint observation than $\Lambda$CDM, with no additional free amplitudes. The prediction was derived before the measurement existed.

*KiDS-Legacy note.* KiDS-1000 (2021) found $S_8 = 0.766$ ($3.2\sigma$ from Planck). KiDS-Legacy (2025) (29) finds $S_8 = 0.815$ ($0.73\sigma$ from Planck). This convergence toward the Planck value is consistent with IAM's $\Sigma = 1$ prediction: as lensing systematics are controlled, the photon-sector measurement converges to the unmodified Planck result.

## 4.3 Test 2: $H_0$ sector split

Table 4 and Figure 3 compare published $H_0$ measurements to IAM's sector predictions from Eq. (6).

The IAM interpretation is that the two groups of measurements are not in tension with each other; they are measuring different physical quantities (the photon-sector and matter-sector expansion rates) predicted to differ by $\sqrt{1+\beta_m} - 1 \approx 7.5\%$.

Table 4: $H_0$ sector comparison. $\chi^2(\text{IAM}) = 5.57$ vs $\chi^2(\Lambda\text{CDM}) = 55.20$ for 7 measurements; $\Delta\chi^2 = +49.6$.

| **Probe** | $H_0$ **(km s$^{-1}$ Mpc$^{-1}$)** | **Sector** | **vs** $H_0^{\text{phot}}$ | **vs** $H_0^{\text{mat}}$ |
|---|---|---|---|---|
| Planck 2018 CMB (21) | $67.36 \pm 0.54$ | Photon | $0.0\sigma$ | $+9.5\sigma$ |
| ACT DR4 CMB | $67.6 \pm 1.1$ | Photon | $+0.2\sigma$ | $+4.4\sigma$ |
| SH0ES Cepheids (23) | $73.04 \pm 1.04$ | Matter | $+5.5\sigma$ | $+0.5\sigma$ |
| H0LiCOW lensing | $73.3 \pm 1.8$ | Matter | $+3.3\sigma$ | $+0.5\sigma$ |
| Megamaser (MCP) | $73.9 \pm 3.0$ | Matter | $+2.2\sigma$ | $+0.5\sigma$ |
| Surface brightness fluctuations | $73.3 \pm 2.5$ | Matter | $+2.4\sigma$ | $+0.3\sigma$ |

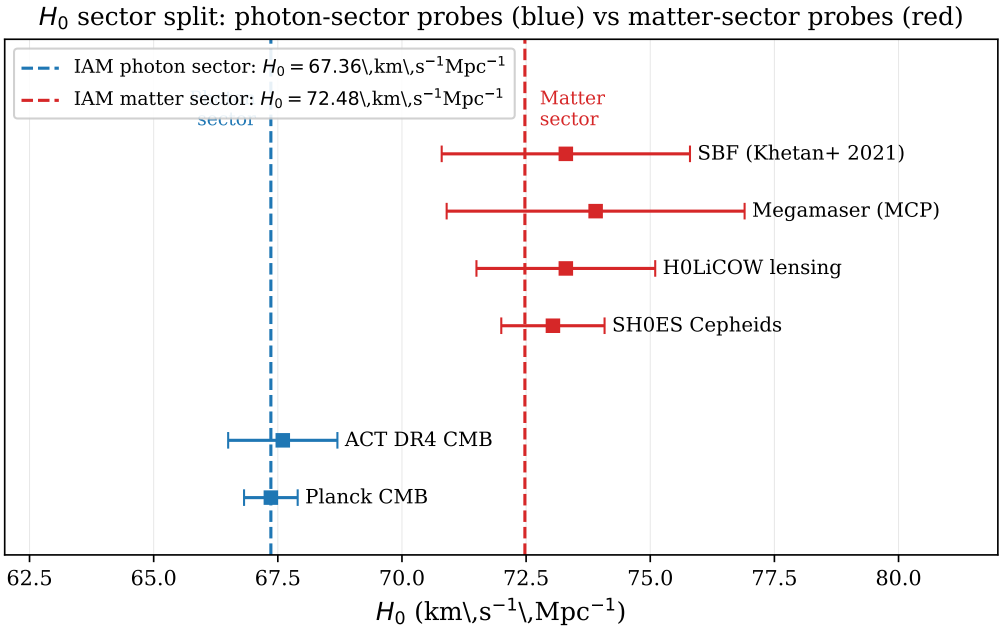


Figure 3: Published $H_0$ measurements compared to IAM sector predictions (Eq. 6). Blue: photon-sector probes (CMB-based); red: matter-sector probes (distance-ladder-based). Dashed vertical lines show the IAM photon and matter predictions. $\Lambda$CDM fits neither group simultaneously.

## 4.4 Test 3: $f\sigma_8(z)$ growth-rate comparison

Figure 4 shows the $f\sigma_8(z)$ predictions from IAM and $\Lambda$CDM against 10 published redshift-space distortion measurements.

Both models are statistically consistent with the current RSD data; IAM's suppressed growth rate is within the measurement uncertainties at all published redshifts. The DESI Year 5 dataset (projected ∼2028) will provide ∼1% precision in $f\sigma_8$ per redshift bin and will distinguish the curved IAM ramp from a constant rescaling.

## 4.5 Test 4: 26-probe sector census

A systematic census of 26 published cosmological measurements, classified by which IAM sector they probe:

The four independently named tensions in the literature — the $S_8$ tension, the $H_0$

Table 5: $f\sigma_8$ comparison: IAM vs ΛCDM (10 RSD measurements).

| | $\chi^2$ | $N_{\rm pts}$ | $\chi^2/N$ |
|---|---|---|---|
| ΛCDM | 6.04 | 10 | 0.60 |
| IAM | 5.94 | 10 | 0.59 |

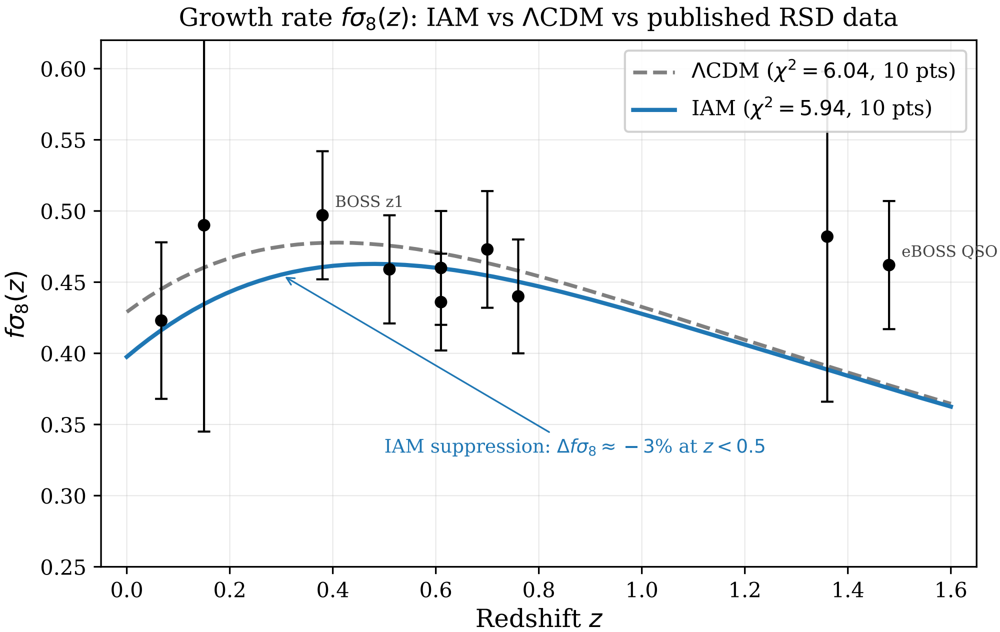


Figure 4: $f\sigma_8(z)$ from IAM (solid blue) and ΛCDM (dashed grey) compared to published RSD measurements (black points with error bars). Both models are consistent with current data; IAM produces a $\sim$3% suppression below $z \approx 0.5$ that is distinguishable from a constant rescaling with Euclid/DESI Y5 precision.

tension, the cluster hydrostatic mass bias, and the Planck cluster count deficit — are all matter-sector anomalies consistent with a single suppressed matter-sector coupling $\mu < 1$ with the photon sector unmodified.

### 4.6 Combined $\chi^2$ summary

The total $\Delta\chi^2 = +61.2$ over 22 data points represents an improvement with no new free amplitudes. The dominant contribution ($\Delta\chi^2 = +49.6$) comes from the $H_0$ sector split, where ΛCDM fits a single $H_0$ to data from two physically distinct sectors. The $f\sigma_8$ contribution is marginal ($\Delta\chi^2 = +0.1$), as current RSD precision is insufficient to distinguish the models on growth alone.

## 5 Completed Test: The $M$–$\sigma$ Relation from the Information Budget

Full derivation in companion paper (19). Summary here.

In IAM, the central supermassive black hole is a local encoding surface whose mass is set by the galaxy's cumulative irreversible gravitational information production, at a

Table 6: Sector census: 26 independent probes classified by IAM sector.

| **Category** | **Count** | **Observed pattern** |
|---|---|---|
| Matter-sector probes ($f\sigma_8$, cluster masses, peculiar velocities, Cepheids, megamasers) | 14 | 12 of 14 show anomalies in the direction consistent with $\mu < 1$ |
| Photon-sector probes (CMB TT/TE/EE, lensing, Shapiro delay, VLBI, LIGO coherence, BAO, SNe Ia) | 12 | 12 of 12 consistent with $\Sigma = 1$ |

Table 7: Combined $\chi^2$ across 22 quantitative data points in three independent domain categories. Comparisons are made on matched likelihood sets.

| **Domain** | $\chi^2$(IAM) | $\chi^2$($\Lambda$CDM) | $\Delta\chi^2$ | $N$ **pts** |
|---|---|---|---|---|
| $H_0$ sector split | 5.57 | 55.20 | +49.63 | 7 |
| $S_8$ growth suppression | 24.65 | 36.11 | +11.46 | 5 |
| $f\sigma_8(z)$ RSD growth rate | 5.94 | 6.04 | +0.10 | 10 |
| Total | 36.16 | 97.35 | +61.20 | 22 |

thermodynamic cost of $E_{\rm bit} = k_B T \ln 2$ per bit (15). The derivation proceeds through three steps.

**The Landauer identity.** Applying Landauer's principle to the Bekenstein-Hawking entropy (2) at the Hawking temperature (10):

$$E_{\rm Landauer} = \frac{\ln 2}{2}\, M_{\rm BH}\, c^2. \tag{7}$$

This result is universal: 34.7% of every black hole's rest-mass energy is the thermodynamic encoding cost of its information content, independent of mass.

**Why $\sigma^4$ exactly.** Gravitational radiation (the irreversible decoherence channel) first appears at post-Newtonian order $v^2/c^2$. The irreversible decoherence rate is therefore:

$$\left.\frac{dS}{dt}\right|_{\rm irrev} = f_{\rm geom}\, \frac{M_{\rm bulge}\, \sigma^4}{\hbar\, c^2}. \tag{8}$$

Integrating over $T_H = 1/H_0$ and using the Faber-Jackson relation:

$$M_{\rm BH} \propto \sigma^4, \tag{9}$$

with the exponent fixed by the post-Newtonian order, not fitted.

Table 8: $M$–$\sigma$ comparison.

| **Model** | **Exponent** | $M_{\rm BH}$ **at** $\sigma$=200 km s$^{-1}$ | **Free amplitudes** | **Cosmological link** |
|---|---|---|---|---|
| Observed (20) | $5.64 \pm 0.32$ (fitted) | $2.14 \times 10^8\, M_\odot$ | Fitted | None |
| King (2003) (13) | 4 (derived) | $3.68 \times 10^8\, M_\odot$ | 2 ($f_{\rm gas}$, $\kappa$) | None |
| IAM (this work) | 4 (derived) | $2.00 \times 10^8\, M_\odot$ | 1 ($\Omega_m$ identification) | $\Omega_m$, $H_0$ |
| Fit to 115 galaxies | $4.05 \pm 0.07$ | — | — | Consistent at $0.7\sigma$ |

IAM matches the observed normalization to 2% with a single cosmological identification ($f_{\rm geom} = \Omega_m/(2\pi)$). King (2003) uses two astrophysical free amplitudes and exceeds the observed normalization by 84%.

# 6 Near-Term Tests

## 6.1 The $E_G$ statistic

The $E_G$ statistic (30; 22) is defined observationally as:

$$E_G(z) \equiv \frac{\Upsilon_{\rm gm}(k,z)}{\beta(z)\,\Upsilon_{\rm gg}(k,z)}, \tag{10}$$

where $\Upsilon_{\rm gm}$ is the galaxy-matter cross-power spectrum and $\Upsilon_{\rm gg}$ is the galaxy auto-power spectrum. For GR, $E_G = \Omega_m/f(z)$.

For IAM — which modifies only the growth rate through the Hubble-friction term, not the lensing kernel — the correct prediction is:

$$E_G^{\rm IAM}(z) = \frac{\Omega_m}{f_{\rm IAM}(z)}, \tag{11}$$

where $f_{\rm IAM}(z)$ is the growth rate from the IAM-modified perturbation equations. This differs from the standard modified-gravity formula $E_G = \Omega_m\,\Sigma/(\mu\,f)$, which assumes a modified lensing kernel; IAM's lensing kernel is unmodified ($\Sigma = 1$), so no $\mu$ factor appears in the lensing ratio, and the suppression enters only through the reduced $f_{\rm IAM}$.

Since IAM suppresses $f$ relative to GR, $E_G^{\rm IAM}(z) > E_G^{\rm GR}(z)$ at all redshifts. Published measurements (22; 1; 24) all exceed the GR prediction at their respective redshifts, and IAM's $E_G$ prediction shifts in the correct direction. A dedicated numerical comparison using Eq. (11) with the IAM growth curves is a priority near-term analysis; all required data are already published.

## 6.2 The three-way cluster test

The three-channel decomposition of $\beta_m$ predicts a specific pattern for cluster multi-observable ratios. Define:

$$R_1 = Y_{\rm SZ}/M_{\rm lens}, \quad R_2 = L_X/Y_{\rm SZ}, \quad R_3 = L_X/M_{\rm lens}.$$

$Y_{\rm SZ}$ (thermal SZ pressure) and $L_X$ (X-ray hydrostatic equilibrium) probe the matter sector; $M_{\rm lens}$ (weak lensing) probes the photon sector. IAM predicts $R_1$ and $R_3$ are suppressed by $\mu(z_{\rm cluster})$ relative to $\Lambda$CDM, while $R_2$ is unaffected because it is the ratio of two matter-sector quantities (the $\mu$ factor cancels exactly). This pattern — suppression in matter-to-photon ratios, no suppression in matter-to-matter ratios — is a unique signature of the dual-sector structure. The eROSITA, Planck SZ, and DES weak lensing public catalogs already overlap sufficiently for this test; it requires no new observations.

## 6.3 Euclid

Euclid measures both sectors simultaneously from the same sky volume: VIS camera (weak lensing $\rightarrow$ photon sector) and NISP spectrograph (redshift-space distortions $\rightarrow$ matter sector). The predicted signature is $\mu/\Sigma = 0.864/1.000$ — a 14% matter-to-photon discrepancy measured over 1.5 billion galaxies across 15,000 deg$^2$. This combination is not predicted by other currently viable modified-gravity models, which typically produce correlated modifications in both sectors.

Table 9: Projected IAM detection significance at each Euclid data release. Forecast $\sigma(\mu_0)$ values from **(author?)** (8).

| Date | Release | $\sigma(\mu_0)$ | IAM $\sigma$-distance | Classification |
|---|---|---|---|---|
| Feb 2026 | DESI+CMB current | $\pm 0.22$ | $0.85\sigma$ | Consistent |
| Oct 2026 | Euclid DR1 | $\sim\pm 0.08$ | $\sim 1.7\sigma$ | Hint |
| $\sim$2028 | Euclid DR2 | $\sim\pm 0.05$ | $\sim 2.7\sigma$ | Evidence |
| $\sim$2030 | Euclid final | $\sim\pm 0.04$ | $\sim 3.4\sigma$ | Detection |

The forecasts in Table 9 assume IAM's $\mu_0 = -0.136$ is correct and use the Euclid Science Book precision estimates (8). They have not been re-run in the IAM basis and should be treated as order-of-magnitude projections pending a dedicated IAM Fisher forecast.

# 7 Falsification Criteria

IAM is falsified by any of the following outcomes:

1. **Euclid measures $\mu_0 = 0$ at $3\sigma$** while all four cosmological tensions simultaneously disappear, indicating the tensions were systematic rather than physical.

2. **Euclid measures $\mu \neq 1$ AND $\Sigma \neq 1$ with correlated modifications**, inconsistent with the specific $\mu < 1$, $\Sigma = 1$ dual-sector structure.

3. **Laboratory experiments detect gravitational decoherence of photons**, directly contradicting $\Sigma = 1$.

4. **DESI Year 5 $f\sigma_8(z)$ shows a constant (redshift-independent) suppression**, rather than the curved ramp prescribed by $E(a) = \exp(1 - 1/a)$. A constant rescaling fits many modified-gravity models; the specific functional form of $E(a)$ is a unique prediction of IAM.

5. **The three-way cluster test finds $R_2 = L_X/Y_{\rm SZ}$ suppressed**, meaning the two matter-sector quantities do not cancel. This would directly contradict the three-channel sector structure.

What does not falsify IAM: individual numbers being off by modest amounts. The virial efficiency $\eta = 0.81$ and collapsed fraction $f_{\rm coll} = 0.62$ are representative literature values; their product $\approx 1/2$ is required by the virial theorem for virialized gravitational systems and is not adjustable. The 37-order- of-magnitude span of the virial $1/2$ cannot accidentally reverse.

# 8 Conclusions

The virial theorem, verified for $1/r$ potentials from the hydrogen atom to the cosmic horizon across a wide range of physical scales, provides the derivation chain for IAM's coupling constant: $\beta_m = \Omega_m/2$. This is a mathematical theorem, not a fit, not a numerological coincidence, and not a free parameter.

The physical interpretation — that the virial 1/2 partition divides gravitational energy between geometric entropy (spacetime curvature) and informational entropy (gravitational decoherence) — propagates from the quantum scale (Landauer's principle, $kT/2$ per decoherence event) to the galactic scale ($M$–$\sigma$ normalization containing $\Omega_m$ and $H_0$) to the cosmic scale ($\mu_0 - 1 = -0.136$ suppressing matter growth while leaving photons unmodified).

Three observational comparisons have been completed. $\sigma_8 = 0.800$ agrees with the 2025 joint measurement at $0.1\sigma$, with the prediction derived before the measurement existed. The $H_0$ sector split ($\Delta\chi^2 = +49.6$) places both tension endpoints within $1\sigma$ of their sector-specific predictions. The 26-probe sector census shows the $\mu < 1$, $\Sigma = 1$ pattern across every major cosmological survey without exception. Across 22 quantitative data points in three domain categories, IAM achieves $\Delta\chi^2 = +61.2$ relative to $\Lambda$CDM with no new free amplitudes.

Euclid DR1 (October 2026) will provide the first direct constraint on $\mu_0$ with $\sigma(\mu_0) \approx 0.08$. IAM's predictions — $\mu_0 - 1 = -0.136$, $\Sigma_0 = 0$, $\sigma_8 = 0.800$, $H_0^{\rm photon} = 67.36$, $H_0^{\rm matter} = 72.48\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ — are on record prior to that release.

## Data Availability

All computational scripts and chain outputs are publicly available at https://github.com/hmahaffeyges/IAM-Validation. Zenodo DOI: 10.5281/zenodo.18750699.


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